

KALEB BRUWER

Cape Town, South Africa

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SUMMARY

Software Engineer with experience across the full software development life cycle, including design, implementation, testing and maintenance. Further experience in compiler technology, program analysis, and formal methods, including the design of semantic analysis pipelines, intermediate representations, and applying automated theorem proving to software verification.

EXPERIENCE

Karoo Compute Solutions

Cape Town, South Africa

Research Software Engineer

May 2025 - Present

- Conducted research and development into automated and verified C to Rust translation
- Defined IRs and lifting stages using Rust to reshape code into the form needed for formal reasoning, while preserving semantics
- Automatically converted the final IR into SMT-LIB form, enabling the Z3 theorem prover to formally verify program equivalence
- Implemented a Cargo test suite consisting of both unit tests and end-to-end tests. This regularly prevented regressions as the product underwent iterative development
- Created a VSCode plugin and Command Line Interface (CLI), which improved user experience and eased LLM integration
- Standardized the development environment and build system with Docker-based CI/CD

EPI-USE Labs / EPI-USE Africa

Pretoria, South Africa

Software Engineer

Jan 2023 - Apr 2025

- Full-stack maintenance and development on FLOW, a Time and Attendance product, with an ABAP backend and a UI5/JavaScript frontend
- Spearheaded development of an in-house REST OData framework to eliminate thousands of lines of boilerplate code
- Introduced a bespoke workflow engine, with a Domain Specific Language (DSL) allowing workflows to be easily customized to each client's needs

EDUCATION

University of Pretoria

2022

BScHons in Computer Science

University of Pretoria

2019 - 2021

BSc in Computer Science

SOFTWARE DEVELOPMENT SKILLS

Languages: Rust, C/C++, SQL, ABAP, Java, JavaScript, Python

Technologies: Git, Linux, Docker, GitHub Actions, GDB

Formal Methods Tools: Z3, SMT-LIB.

ACHIEVEMENTS AND AWARDS

SANReN Cyber Security Challenge

2021

Capture the Flag Challenge between teams from various Southern African universities; we placed second overall

ASSA South African Mathematics Team Competition

2017, 2018

Qualified for and competed as part of the Gauteng North mathematics team; in 2018 we placed third nationally

PROJECTS

- rprop:** Propositional logic in Rust's type system 2026
- Architected a procedural-macro Rust library, published on crates.io
 - Made use of procedural macros to allow logical propositions, claims and proofs to be written in a simple and intuitive syntax
 - Leveraged Rust's type system to verify proofs at compile time
- Spruce Goose:** A C++ Minecraft server 2020 - 2022
- Implemented a Minecraft server in C++, compatible with unmodified clients, available on [GitHub](https://github.com)
 - Implemented TCP-socket networking, mirroring the official Minecraft protocol, enabling standard client connections
 - Multithreading and inter-thread communication
 - Built data structures for efficiently handling 3D spatial data, to improve scalability
- Unison:** Smart Contract Verifier 2021
- Final-year student project leading a team of 5, formed part of Computer Science degree
 - Created a DeFi escrow service, where we used blockchain/smart contract technology to track user agreements in a decentralized manner, where anonymous juries settled disputes